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CHAPTER 34

Body Image and Binge-Eating Disorder

JOSHUA I. HRABOSKY

INTRODUCTION

Binge-eating disorder (BED), a research category in the fourth edition of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-IV), is defined by recurrent binge eating without the presence of maladaptive compensatory weight control behaviors characteristic of bulimia nervosa (BN). Binge eating is defined as eating an unusually large amount of food given the circumstances while experiencing a subjective sense of loss of control during the eating episode. Binge-eating episodes are further characterized by eating more rapidly than normal; eating beyond a sense of fullness; eating despite the absence of hunger; hiding the extent to which one is eating; and/or experiencing disgust, depression, or guilt associated with one's eating. In its current state, the DSM-IV requires binge eating must occur, on average, at least 2 days per week for no less than 6 months; however, the recently proposed revisions for the DSM-5 require binge episodes must occur, on average, at least once per week for no less than 3 months.

BED has received support as a prevalent and chronic psychiatric disorder, especially in comparison to BN and anorexia nervosa (AN). Binge eating is associated with numerous additional clinical problems, including depressive symptomatology, low self-esteem, and poor quality of life. Research has further found high rates of psychiatric comorbidity in individuals with BED compared to individuals without BED. BED is predominantly associated with overweight and obesity, and is, consequently, associated with greater risk of medical morbidities and mortality. (See Chapter 21 for a discussion of adulthood obesity and body image.) In fact, obese patients with BED suffer from greater health dissatisfaction, poorer health-related quality of life, and higher rates of major medical

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disorders (e.g., hypertension, diabetes, cardiac problems) than obese patients without BED. Further, BED is associated with increased health care utilization, making it a major target for treatment and prevention within the larger subset of obese individuals.

SHAPE AND WEIGHT OVERVALUATION

Research on the presence and mediating role of body image in BED has historically focused on the degree of *concern* with one's body weight and/or shape. This has been most commonly defined by the Eating Disorder Examination's (EDE) shape and weight concern subscales, which comprise items purportedly assessing a range of constructs, including fear of weight gain, preoccupation with shape or weight, dissatisfaction with shape and weight, and discomfort seeing one's body. The EDE, both in its interview and self-report (EDE-Q) forms, has been the most often-used instrument for measuring the psychopathology of BED. Thus, its shape and weight concern subscales have historically been the primary source of knowledge about body image in BED. Individuals with BED have been found to report significantly greater degrees of shape and weight concern than both overweight and normal-weight individuals without BED, while similar in degree to individuals with AN and BN.

Within the past decade and, to a greater degree the past couple of years, the focus of research on understanding body image in BED has shifted to the excessive influence of shape or weight on one's self-evaluation, or *overvaluation*. Shape/weight overvaluation has been defined as the importance one places on her or his physical appearance (specifically, one's body shape and weight) in defining her- or himself. Overvaluation, or *investment*, is associated with, yet distinct from, the degree of satisfaction or dissatisfaction with one's appearance. In fact, although many may endorse dissatisfaction with their body shape or weight, only a portion will report that the appearance of their shape or weight significantly impacts their self-worth. Moreover, as Masheb and Grilo found in a longitudinal study of body image in BED, shape/weight overvaluation appears to be more stable and more closely associated with self-esteem than body dissatisfaction.

Shape/weight overvaluation has been found to be of critical importance in characterizing and maintaining AN and BN, as well as the more prevalent eating disorder not otherwise specified (EDNOS). Consequently, this construct (i.e., "undue influence of body weight or shape on self-evaluation") is currently included in the DSM-IV (p. 545) as a diagnostic criterion for both AN and BN. Yet, debate continues regarding the specific diagnostic criteria for the BED diagnosis, including whether, like AN and BN, it too should be characterized by overvaluation of shape and weight.

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Recent research has supported shape/weight overvaluation as a distinguishing clinical feature within patients with BED. Hrabosky, Masheb, and colleagues found 58% of a sample of 399 treatment-seeking patients with BED endorsed clinical overvaluation. This subset of patients reported greater eating-related and general psychopathology than those experiencing subclinical overvaluation. Several recent studies including clinical and community-based samples have replicated, as well as extended, the findings of Hrabosky and colleagues, comparing these two BED subgroups with overweight individuals without BED, individuals with BN or subthreshold BN, and psychiatric controls. These studies found individuals with BED, regardless of degree of overvaluation, suffer from greater eating pathology than overweight individuals and psychiatric controls. Further, patients with BED endorsing clinical overvaluation differed little from patients with BN on measures of eating pathology and general psychological disturbance. High levels of shape/weight overvaluation in BED is associated with greater body dissatisfaction, eating pathology, depressive symptomatology, social dysfunction, and rates of lifetime psychiatric comorbidity; poorer self-esteem and quality of life; and increased health care utilization. Many of these associations persist after controlling for depression/negative affect. Interestingly, although patients with BED experiencing clinical overvaluation endorsed greater distress associated with binge eating, several studies have found no differences between patients with clinical overevaluation and those with subclinical overvaluation in the frequency of binge eating. Nor do these groups differ significantly in body mass index.

These findings have resulted in a call for the addition of a diagnostic *specifier* for BED in DSM-5, identifying a distinct subgroup of individuals who suffer from a key cognitive feature commonly associated with clinically relevant psychological and behavioral disturbances. Whereas some have suggested that shape/weight overvaluation should be included as a diagnostic *criterion*, doing so would exclude a substantial proportion of individuals who engage in recurrent binge eating and suffer from clinically significant behavioral and psychosocial problems. Nonetheless, the existing data support the clinical significance of shape/weight overvaluation in BED.

BODY IMAGE COPING STRATEGIES IN BED

Although shape/weight overvaluation is considered core to the psychopathology of disordered eating, body image coping strategies are hypothesized to be its behavioral expression. Individuals suffering from body image distress engage in a range of mental or behavioral strategies to cope with various threatening situations. Such maladaptive coping practices include body checking, body avoidance, appearance correcting, and camouflaging, and serve multiple purposes. Mountford and

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colleagues found female patients with an eating disorder, including a subset of patients with BED, engage in body checking in an effort to decrease anxiety through reassurance, avoid negative consequences that may occur without body checking, and control dietary intake and weight. Body checking, like other self-regulating strategies, allow the individual to escape or reduce, albeit momentarily, the psychological or emotional distress associated with the individual's physical appearance, consequently maintaining negative body image attitudes through negative reinforcement.

Reas and colleagues found self-report items assessing body checking and body avoidance in a sample of treatment-seeking patients with BED were significantly but differentially associated with disordered eating behaviors. Checking behavior was positively associated with dietary restraint, whereas body avoidance was positively associated with binge eating. Reas and colleagues hypothesized that patients' fluctuating feelings of loss of control are associated with the appearance-managing strategy they exercise. Mountford and colleagues similarly found a significant correlation between dietary restraint and appearance checking. They further discovered that restraint was positively associated with the degree to which individuals believe body checking helps control their dietary intake and weight. Despite temporarily reducing anxiety through reassurance, body checking purportedly magnifies perceived imperfections, motivating the individual to engage in further dietary restriction. On the other hand, efforts in avoiding appearance-related triggers, thoughts, or emotions may include binge eating. Disinhibited eating has been posited as a means of experiential avoidance (e.g., attempts to ignore body image-related thoughts or triggering situations).

Patients with BED particularly engage in maladaptive coping strategies for managing body image distress. Reas and colleagues found a substantial subset (57%) of patients with BED endorsed pinching areas of their body to check for fatness, whereas an even larger subset (89%) reported often avoiding wearing clothes that make them aware of the shape of their body. Comparing the body image experiences of treatment-seeking obese patients with and without BED, Hrabosky, Cash, and colleagues found patients with BED were significantly more likely than their non-BED counterparts to cope with body image threats by engaging in body avoidance, experiential avoidance, appearance-fixing behaviors, and self-regulation through eating.

STABILITY OF BODY IMAGE OVER TIME

BED has been supported as a stable disorder, comparable in duration and chronicity to other eating disorders (i.e., AN, BN, and EDNOS). Although research has focused primarily on the duration and resolution of eating disorder features characteristic of BED and central to its diagnosis, body image problems also appear to persist. Striegel-Moore and colleagues examined the course of body image

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before, at the time of, and prior to the time of diagnosis of an eating disorder in adolescent females. The authors found that, similar to young women diagnosed with BN, women with BED endorsed significantly greater weight dissatisfaction than healthy comparisons at all time points ranging from 2 years prior to diagnosis of BED to 2 years after diagnosis. Moreover, the data suggested these group differences increased over time, reflecting worsening body image in BED.

To gain insight into the long-term course of BED and BN, Fichter and colleagues compared these groups of female patients, along with healthy controls, at presentation for treatment and at 12-year follow-up. Although patients with both BED and BN displayed significant reductions over time in self-report measures of body dissatisfaction and drive for thinness, patients with BED suffered from poorer body image than their BN counterparts at both time points. In addition, higher body dissatisfaction was predictive of general severity of BED at 12 months follow-up. Most interesting, however, was the authors' finding that, despite the resolution of eating disorder symptoms, a subset of patients with a previous diagnosis of BED continued to endorse a thinner body shape ideal than healthy comparisons. Cognitive processes, such as body image may, therefore, be more chronic and persisting than the problematic behaviors with which they are correlated.

BODY IMAGE AND TREATMENT OF BED

Body Image and Treatment Outcome

Specialized treatments, such as cognitive-behavioral therapy (CBT) and interpersonal psychotherapy (IPT) have received empirical support as efficacious interventions for producing statistically and clinically significant reductions in, or full remission of, binge eating. Yet, a sizeable proportion of patients do not fully respond to treatment. Efforts have begun toward identifying patient predictors of treatment outcome and the mechanisms by which treatments result in behavioral change.

Both shape/weight concern and shape/weight overvaluation have received support as predictors of treatment outcome of BED. Dingemans and colleagues found that full remission of binge eating following group CBT was fully mediated by changes in weight concerns from pre- to post-treatment and marginally by changes in shape concerns. Weight and shape concerns did not, however, predict binge-eating status 1 year post-treatment. Hilbert and colleagues examined putative pre- and midtreatment predictors of post- and follow-up response to CBT and IPT. The authors found that, among those individuals with low (compared to high) interpersonal problems, greater shape/weight

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concern at both pre- and midtreatment was predictive of posttreatment nonresponse. Similar to the previous study, Hilbert and colleagues found shape/weight concern was not predictive of binge-eating status 1 year following treatment's end.

In addition to being a significant correlate of concurrent psychopathology and dysfunction, shape /weight overvaluation has received support as a predictor of treatment outcome. Masheb and Grilo categorized patients receiving guided self-help CBT or behavioral weight loss (BWL) into clinical and subclinical overvaluation groups based on baseline reports, and found patients endorsing pretreatment shape/weight overvaluation suffered from greater eating pathology at the end of treatment. These analyses were repeated with a sample of patients who received CBT and BWL group therapy and, regardless of treatment condition, baseline overvaluation status was predictive of posttreatment binge-eating frequency.

Treatment of Body Image in BED

Within CBT for BED, the primary aim of treatment for body image disturbances involve identifying and correcting maladaptive assumptions associated with body shape and weight (see Chapter 47 on body image CBT). However, as noted above, cognitive-behavioral formulations of body image in BED have identified behavioral aspects (e.g., body checking and avoidance) in the maintenance of body image disturbance and binge eating. Body exposure techniques have been increasingly implemented within the context of CBT of eating disorders with the aim of addressing both cognitive and behavioral aspects of body image disturbance. Central to body exposure is the in vivo provision of visual feedback, often via a mirror, of one's physical appearance. While the initial goals of body exposure are to correct distorted perception and reduce avoidance, Hilbert and colleagues contend additional purposes include inducing changes in negative body-related schema as well as habituation of negative body-related feelings. In an initial study of the impact of body exposure on mood, appearance-based self-esteem, and negative thinking, Hilbert and colleagues delivered the intervention over two sessions to patients with BED and healthy controls. Results indicated that, while within sessions, patients with BED suffered from increased negative mood and decreased appearance self-esteem, mood significantly improved after follow-up in patients with BED and, regardless of group, appearance self-esteem increased. Further, negative mood and frequencies of negative cognitions reduced and appearance self-esteem increased by the time of the second exposure session. The authors concluded, despite decreases in the frequencies of negative cognitions, changes in cognitive content may require further body exposure and/or cognitive restructuring.

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Hilbert and Tuschen-Caffier extended the results of this study, randomly assigning women with BED to receive CBT with body exposure or CBT with cognitive restructuring. The two treatment conditions resulted in statistically similar and substantial improvements in shape/weight concerns and body dissatisfaction at posttreatment and were maintained at 4-months posttreatment. Similarly, negative body image thoughts decreased in frequency by follow-up. Although the study was limited by small sample sizes (12 patients per group), the authors contend the results suggest that the previously established CBT protocol will not be necessarily enhanced through the alternative use of body exposure rather than cognitive restructuring.

CONCLUSIONS AND FUTURE DIRECTIONS

Despite the absence of a body image-related criterion for BED in the DSMIV, a burgeoning body of research has been directed toward understanding the role of body image in this particular eating disorder. Nonetheless, there are limitations in the existing literature and numerous empirical questions remain. Previous research has relied heavily on psychometrically flawed measures of body image. The proposed subscale structure of the EDE has failed to be replicated, whereas another commonly used measure, the Body Shape Questionnaire, has been criticized for its length and the inclusion of unnecessary items unrelated to its proposed unidimensional construct (i.e., body image preoccupation). Alternatively, more psychometrically sound measures of the multifaceted body image construct should be considered. Using previously validated measures of related, yet discrete, body image dimensions, Hrabosky and colleagues found that obese patients with BED endorsed more severe global and body area-specific body dissatisfaction, greater body-related situational distress and appearance investment (i.e., overvaluation), and poorer body image-related quality of life than their non-BED counterparts.

Further research is necessary to understand the clinical significance of shape/weight overvaluation in BED. While both binge eating and body image disturbance have been supported as correlates of general psychological distress and impaired psychosocial functioning, little is known about the degree to which these two variables uniquely influence distress and dysfunction within BED. Mond and colleagues found shape/weight concern to be a stronger mediator of the influence of obesity on psychosocial functioning than binge eating. Treatments for BED are evaluated in their effectiveness in inducing reduction or remission of binge eating; yet early outcome research with BN suggests that despite remission of behavioral symptoms by the conclusion of treatment, the continued presence of

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shape/weight overvaluation results in increased risk of relapse in behavioral symptoms. Future research should consider examining outcome of shape/weight overvaluation following treatments for BED, as well as whether the continued presence of overvaluation posttreatment predicts relapse.

Despite increased empirical efforts to understand the mechanisms of binge episodes in natural settings (i.e., through the use of ecological momentary assessment [EMA] rather than relying on retrospective self-report measures), no known research has examined reactivity to body image-triggering situations as a precipitant to maladaptive eating in BED. Future research should explore the situational reactivity of patients with BED through EMA, examining the schema-driven cognitive processes in both patients who endorse and deny clinically significant shape/weight overvaluation, as well as the degree to which body image-related thoughts and emotions precipitate binge episodes.

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